

Task: Vision-Based Landing Target Detection

Objective

Design and implement a **basic vision-based landing target detection system** that could run on a companion computer (e.g., Jetson Orin Nano) and support precision landing of a rocket hopper.

The system should detect a landing target from image input and estimate its position relative to the camera.

This task would evaluate your ability to combine computer vision, data processing and system-level thinking in an aerospace context.

It would be a 1.5 week task.

Task Overview

You will build a small pipeline that:

1. Takes image input (dataset or simulated frames)
2. Detects a landing target
3. Outputs the target position in image coordinates.

1. Landing Target Definition

Assume a simple, known landing marker:

- Example:
 - Circular pad
 - “H” marker
 - AprilTag / ArUco marker (optional)

You may:

- Use an existing dataset
- Generate synthetic data
- Or create your own simple dataset

2. Detection System

Implement a method to detect the landing target in each frame.

You may choose:

- Classical CV (OpenCV):
 - thresholding, contours, feature detection
- OR lightweight ML approach:
 - small CNN / pretrained model

Requirements:

It should be robust to:

- noise
- small lighting variations
- moderate scale changes

3. Position Estimation

From the detected target, compute:

- Target center in image coordinates (required)
- Optionally:
 - relative offset from image center
 - approximate distance (if assumptions are made)

4. Real-Time Processing

Simulate a real-time pipeline:

- Process frames at ~10–30 FPS (can be approximate)
- Show detection overlay on images/video

5. System Design

Structure your solution clearly:

- Image input module
- Detection module
- Output / visualization module

Deliverables

1. Code Repository

- Clean and modular code
- Instructions to run it.

2. README

You should explain:

- Approach used (CV vs ML)
- Why you chose it
- Assumptions made

3. Design Note (1–2 pages)

Discuss:

- Trade-offs of classical CV vs ML approaches
- Performance considerations (Jetson constraints)
- Failure cases (e.g., lighting)
- How this would integrate into a real hopper system (camera → Jetson → flight controller)

Final Note

We are not expecting a perfect or production-ready vision system. The focus is on how you go about the task and your awareness of performance constraints in such a system.

We look forward to your approach!